PROJECT-BASED LEARNING

PHASE-II EVALUATION

JEEVANROUTE:Emergency ambulance selection,hospital capability matching and route calculation

Team ID: DSCPP-III-2026-T194

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T194
Semester: 3rd
Section: A, B, C, ML2
Domain: *Core, AI ML*

Team Members

1. *Abhishek Kumar Yadav- 2510011034*
2. *Archit Singh- 2510010859*
3. *Aryan Singh- 2510010893*
4. *Tushar Srivastava- 2510370201*

Mentor

Mr.Kireet Joshi

Interactions completed: 5

Brief Discussion: Introduce the team, project domain, mentor, and responsibilities of the team members.

Phase-I Recap & Feedback Incorporated

Phase-I Status

Proposed a C++-based emergency ambulance and hospital routing system. Finalized the project concept, objectives, modular architecture, and planned data structures for patient prioritization, ambulance selection and hospital matching.

Feedback Received

- *Improve emergency prioritization based on patient severity.*
- *Consider hospital facilities instead of selecting only the nearest hospital.*
- *Improve route selection and divide the system into separate modules.*

Changes Incorporated

- *Planned a priority queue to process critical emergencies first.*
- *Added hospital capability matching.*
- *Adopted graph-based route calculation and a modular C++ design.*

Problem Statement & Finalized Objectives

Problem Statement

Rural emergency cases can be delayed when ambulance availability is limited, the nearest hospital lacks required facilities, or the route is inefficient. JEEVANROUTE is proposed as a decision-support prototype that combines patient priority, ambulance availability, hospital capability and road-network information.

Finalized Objectives

1. *Prioritize emergency patients using a priority queue based on severity.*
2. *Select suitable available ambulances for emergency cases.*
3. *Match hospitals based on required facilities, ICU beds, CT scans, and specialist availability.*
4. *Calculate efficient ambulance routes using graphs and shortest path algorithms.*

Updated System Architecture / Workflow



Major Components

- *Patient Management and Emergency Prioritization*
- *Ambulance Selection and Hospital Capability Matching*
- *Graph-Based Route Calculation and Hospital Pre-Alert*

Changes Since Phase-I

- *Organized the system into separate modules for easier development and integration.*
- *Connected emergency prioritization, ambulance selection, hospital selection, and route calculation into a single workflow.*

Module Design & Functional Requirements

Major Modules

1. *Patient Management and Emergency Prioritization*
2. *Ambulance Management*
3. *Hospital Capability Matching*
4. *Graph-Based Route Calculation*

Functional Requirements

- *Register and search patient records; prioritize emergencies.*
- *Track ambulance availability and select a suitable ambulance.*
- *Match hospitals based on required facilities and specialists.*
- *Calculate efficient routes using graphs and Dijkstra's algorithm.*

Technology Stack & Development Methodology

Technology Stack

- Programming: *C++*
- Development Tool: *Visual Studio Code*
- Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Module Development
4. Integration
5. Testing & Validation

Brief Discussion: C++ supports the implementation of data structures, algorithms, and OOP concepts. File handling provides persistent storage, while GitHub enables team collaboration. The development methodology ensures systematic module development, integration, and testing.

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures in C	*Linked Lists, Priority Queues, Graphs, Breadth-First Search (BFS), Searching and Sorting*	*Patient Management and Route Calculation*
OOPs with C++	*Classes, Objects, Encapsulation, Inheritance and Polymorphism*	*Patient, Ambulance and Hospital Management*

Important: Mention only those concepts that have been **actually implemented** in the project.

Development & Implementation Progress

Module	Status	Work Completed
Patient Management	Completed	*Data structures and record management planned*
Emergency Prioritization	Completed	*Priority queue logic planned*
Ambulance and Hospital Management	In Progress	*Selection and facility matching planned*
Graph-Based Route Calculation	Pending	*Dijkstra-based routing to be implemented*

Overall Project Completion: *70%*

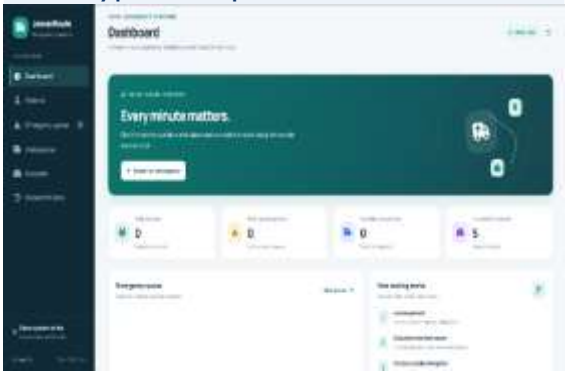
Brief Discussion: Clearly differentiate between completed, in-progress and pending development. Do not present proposed work as completed work.

Working Prototype / Demonstration

Implemented Features

- *Patient record management design*
- *Emergency prioritization using a priority queue*
- *Hospital capability matching*
- *Graph-based route calculation using Dijkstra's algorithm (planned)*

Prototype / Output



Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
Emergency Prioritization	*Critical cases processed first*	*Working*	Pass
Hospital Matching	*Hospital lacking required facilities rejected*	*Working*	Pass
Route Calculation	*Shortest valid route returned*	*Implementation remaining*	*Pending*

Results / Validation

Emergency prioritization and hospital capability matching have been implemented and tested successfully. Graph-based route calculation using Dijkstra's algorithm remains to be implemented and tested. Final system validation will be conducted after integration of all modules.

GitHub Repository & Development Activity

Repository Information

- GitHub Link: https://github.com/Abhishek-co-10/JEEVANROUTE
- README.md: *Updated*
- Contributors: *All 4 Members Added*
- Repository Status: *Updated*
- Branches: *main*

Development Activity

- *Regular commits maintained*
- *Meaningful commit messages*
- *Individual contribution visible*
- *Latest implementation uploaded*
- *Documentation maintained*

Requirement: All team members must be contributors and must **commit individually using their own GitHub accounts**.

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Abhishek Yadav	*Team Lead*	*Repository setup, project architecture and module integration*	*18*
Archit Singh	*Developer*	*Patient data structures and ambulance selection*	*06*
Aryan Singh	*Developer*	*Module interfaces, graph-based route calculation and hospital pre-alert integration*	*06*
Tushar Srivastava	*Developer*	*Emergency assessment and priority queue design*	*05*

Individual Evaluation

Every member should be prepared to explain and demonstrate their own technical contribution to the project.

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- *Managing patient records and emergency priorities.*
- *Integrating independently developed modules.*
- *Handling unavailable ambulances and hospitals lacking required facilities.*

Solutions / Learning

- *Used a priority queue for emergency prioritization.*

Phase-III Roadmap

1. *Implement graph construction and road-distance storage.*
2. *Implement Dijkstra's shortest-path algorithm.*
3. *Connect graph routing with the integrated ambulance and hospital modules.*
4. *Test multiple routes, unreachable hospitals, and edge cases.*
5. *Validate the complete workflow and finalize documentation.*

Expected Outcomes & References

Expected Final Outcomes

- *Complete C++ emergency routing prototype.*
- *Tested emergency prioritization and hospital matching.*
- *Graph-based shortest-route calculaImproved emergency coordination and hospital selection.*
- *Scope for future real-time GPS and traffic integration.*

References

1. *C++ Reference - https://en.cppreference.com/*
2. *GeeksforGeeks - Data Structures and Algorithms*
3. *Thomas H. Cormen et al. - Introduction to Algorithms*
4. *Git Documentation - https://git-scm.com/doc*
5. *JEEVANROUTE GitHub Repository - https://github.com/Abhishek-co-10/JEEVANROUTE*